

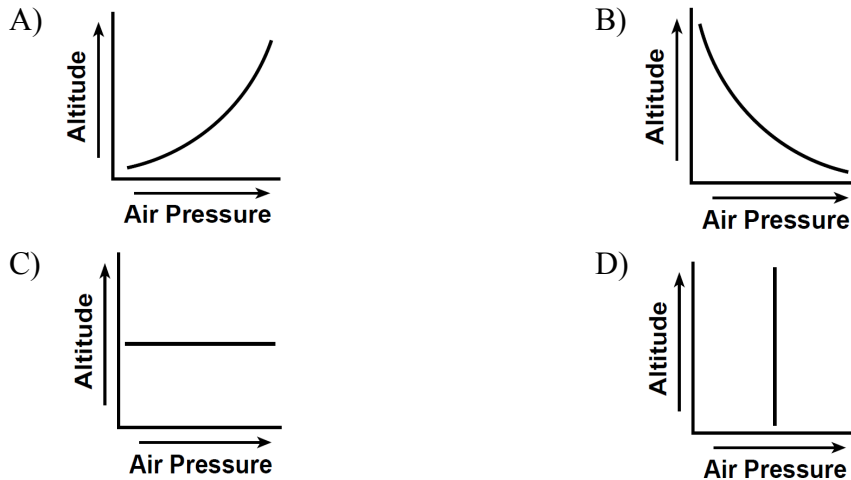
Unstructured Problem Solving

High Pressure and Low Pressure

① The air over the Equator generally rises because the air is

- A) dry and cool with low density
- B) moist and hot with low density
- C) moist and cool with high density
- D) dry and hot with high density

② Which graph best shows the general relationship between air pressure and altitude?



③ The presence of which atmospheric condition will most likely result in clear skies in Binghamton, New York?

- A) high humidity
- B) high-pressure center
- C) increasing wind speed
- D) dewpoint equal to air temperature

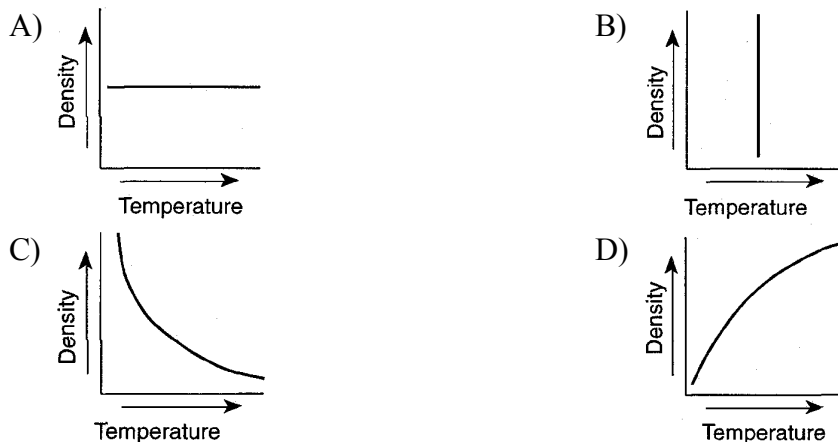
④ As warm, moist air moves into a region, barometric pressure readings in the region will generally

- A) decrease
- B) increase
- C) remain the same

⑤ Air pressure is usually highest when the air is

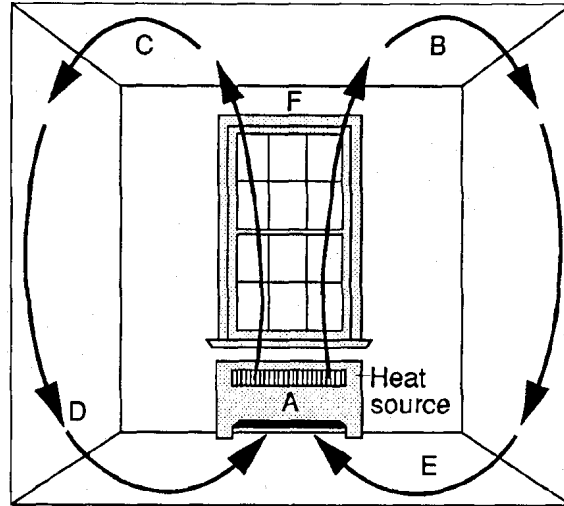
- A) warm and humid
- B) warm and dry
- C) cold and humid
- D) cold and dry

⑥ Which graph best represents the relationship between air temperature and air density in the atmosphere?



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- ⑦ Base your answer to the following question on the diagram below. The diagram shows the pattern of air movement within a closed room.



If the room is considered a model of circulation in the Earth's atmosphere, at which location would air pressure be greatest?

- A) *A* B) *B* C) *E* D) *F*
- ⑧ As the temperature of the atmosphere at a given location increases, the air pressure will most likely
- A) decrease B) increase
C) remain the same
- ⑨ Most of the air in the lower troposphere at the equatorial low-pressure belt is
- A) warm, moist, and rising B) warm, dry, and rising
C) cool, moist, and sinking D) cool, dry, and sinking
- ⑩ Base your answer to the following question on the information below, which describes the past and present climate of Antarctica, and on your knowledge of Earth science.

Antarctica's ice sheet has an average thickness of 6600 feet and holds approximately 70% of Earth's freshwater. Ice layers in Antarctica preserve information about Earth's history. Fossil evidence found in the bedrock of this continent shows that Antarctica was once tropical and is a potential source of untapped natural resources. Antarctica is now a frozen desert with very little snowfall.

Explain why Antarctica's cold climate is responsible for its very low amount of yearly precipitation.